

Weekly Analysis Update: August 14, 2026

Chatura Kuruppu

New Mexico State University
SeaQuest Experiment (E906)

August 20, 2026

- Review of comments from the recent E906 presentation
- Updated cross-section results
- Evaluation of unfolding systematic uncertainties

Review of Comments on Unfolded Cross-Section Results (Part 1)

❶ **Were Messy/Clean corrections applied during unfolding?**

Messy/Clean corrections are accounted for during the acceptance correction phase, which is handled separately from the unfolding process.

❷ **How should the large standard deviations in the ratio above 7 GeV be interpreted?**

Error bars were calculated by rigorously propagating uncertainties, including fully correlated systematic effects (which were previously underestimated).

❸ **We need to define the systematic uncertainty introduced by unfolding. (Proposed method: calculating the systematic uncertainty by varying the number of iterations by one step.)**

Why is Clean unfolding preferred over Messy unfolding?

Using Messy MC for unfolding risks double-counting corrections, as the acceptance correction already incorporates Messy MC effects.

Review of Comments on Unfolded Cross-Section Results (Part 2)

- 4 **Is it possible to use a different number of iterations for Messy versus Clean unfolding?**

Currently, we determine the optimal number of iterations by minimizing the divergence between the mean statistical error and the $\Delta\chi^2$ convergence curves.

- 5 **To determine unfolding systematics, could we adopt the closure test methodology used by Harsha?**

This requires further discussion with the working group.

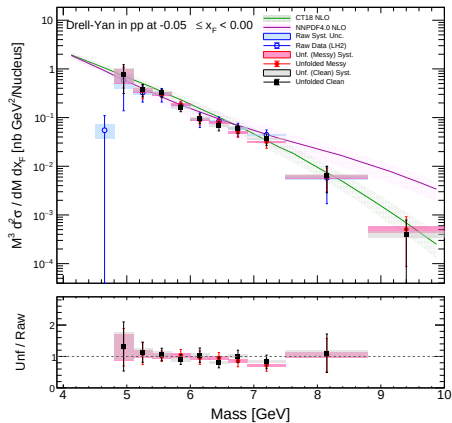
- 6 **Combine overflow and underflow bins, and investigate the required bin widths (they should exceed the detector resolution).**

We plan to broaden the peripheral bins and evaluate the impact on the final cross-section results.

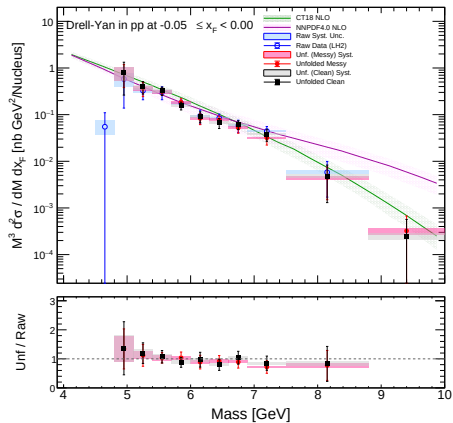
Updated Cross-Section Results

LH2 Target: $-0.05 \leq x_F < 0.00$

Iterations = 3

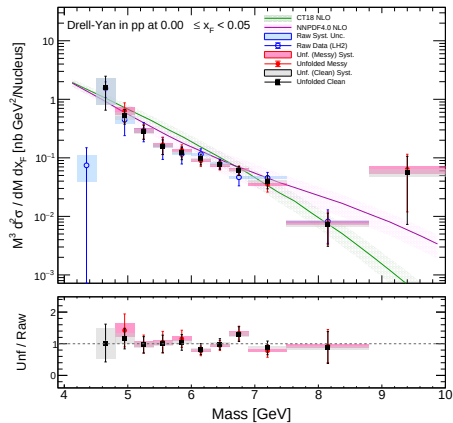


Iterations = 4

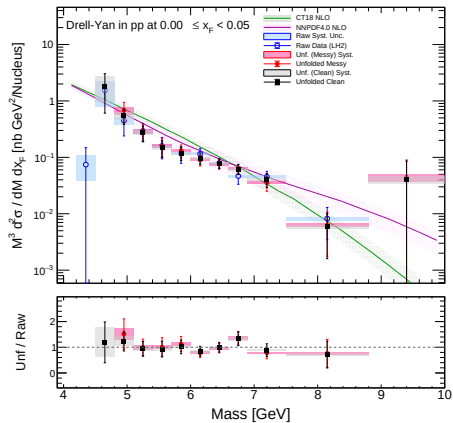


LH2 Target: $0.00 \leq x_F < 0.05$

Iterations = 3

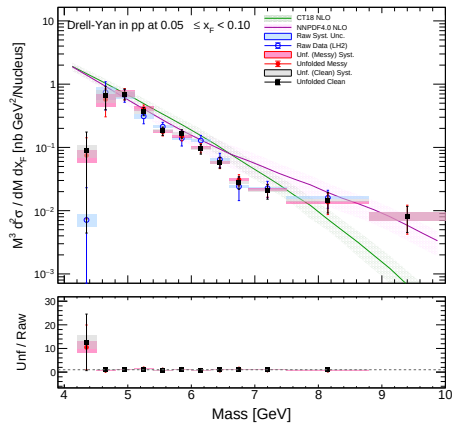


Iterations = 4

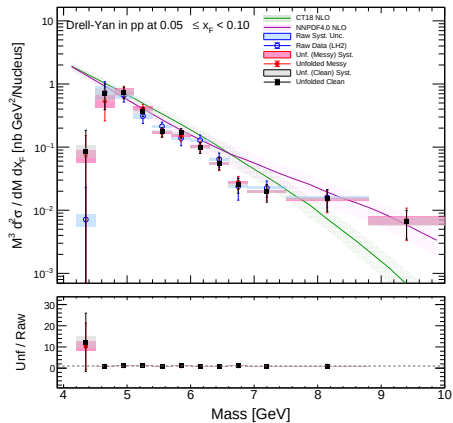


LH2 Target: $0.05 \leq x_F < 0.10$

Iterations = 3

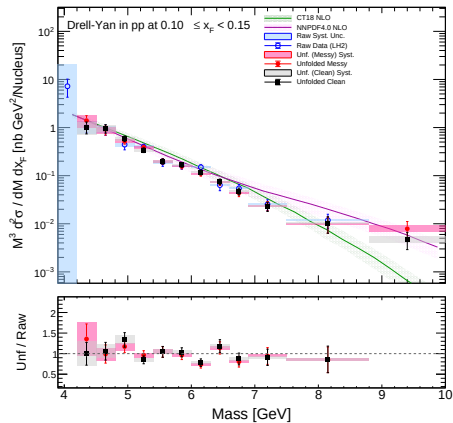


Iterations = 4

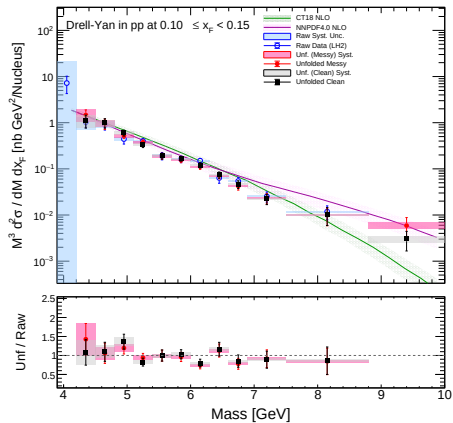


LH2 Target: $0.10 \leq x_F < 0.15$

Iterations = 3

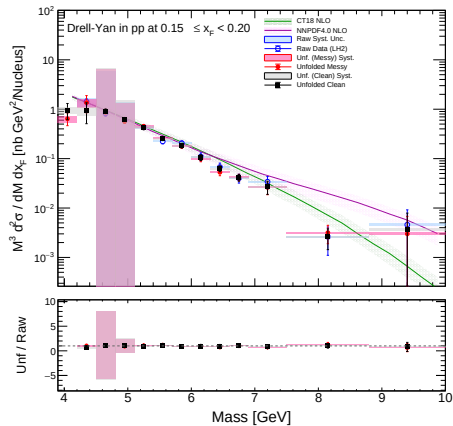


Iterations = 4

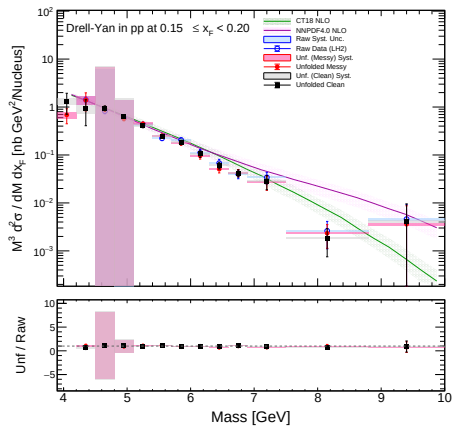


LH2 Target: $0.15 \leq x_F < 0.20$

Iterations = 3

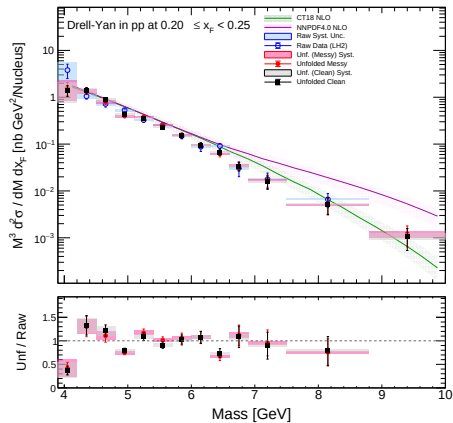


Iterations = 4

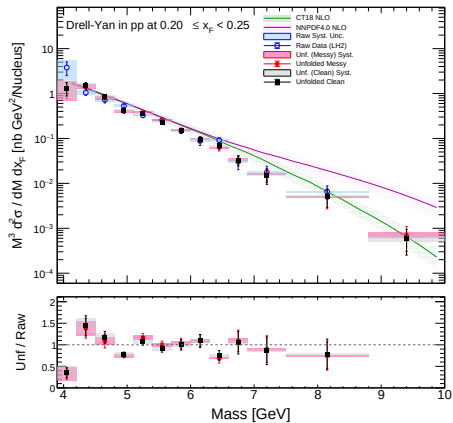


LH2 Target: $0.20 \leq x_F < 0.25$

Iterations = 3

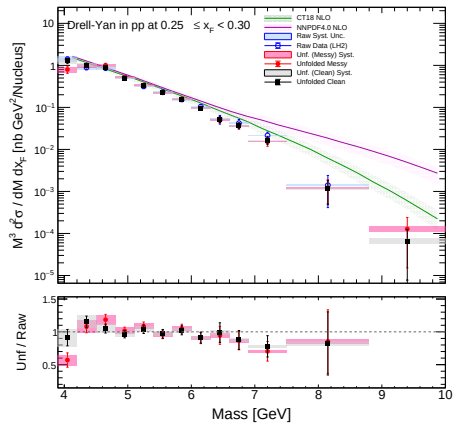


Iterations = 4

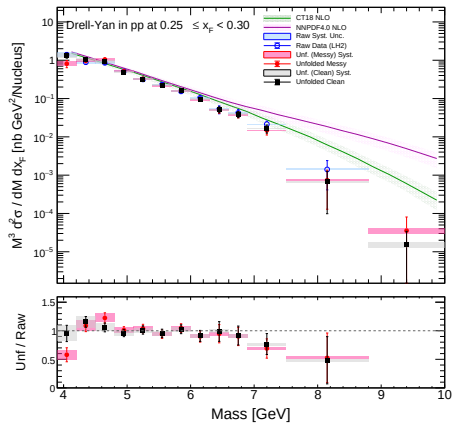


LH2 Target: $0.25 \leq x_F < 0.30$

Iterations = 3

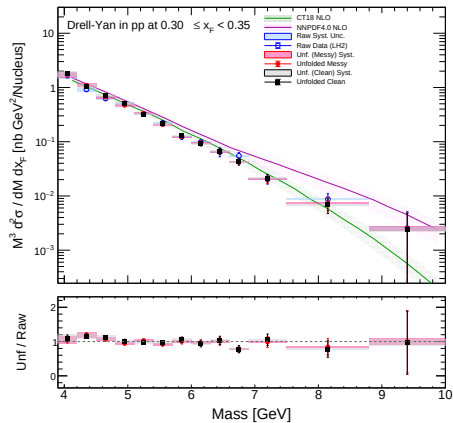


Iterations = 4

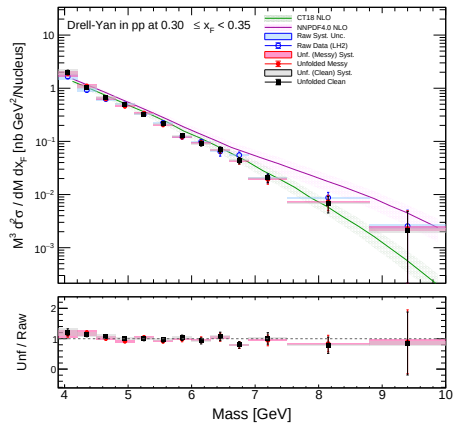


LH2 Target: $0.30 \leq x_F < 0.35$

Iterations = 3

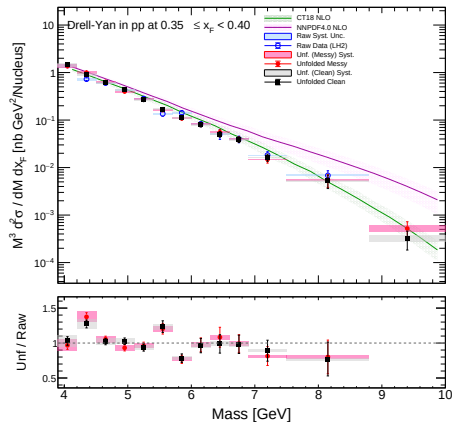


Iterations = 4

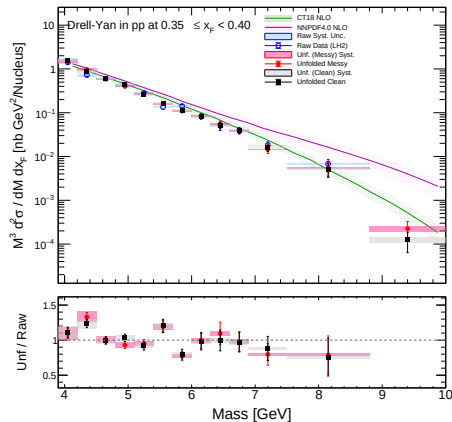


LH2 Target: $0.35 \leq x_F < 0.40$

Iterations = 3

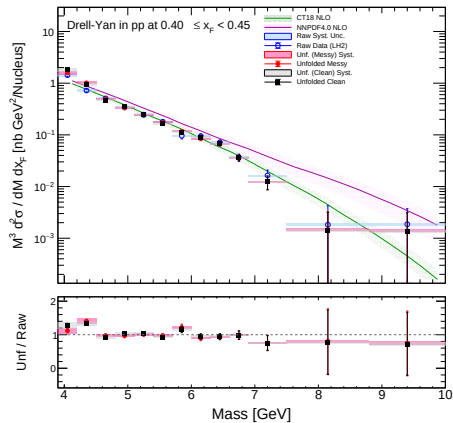


Iterations = 4

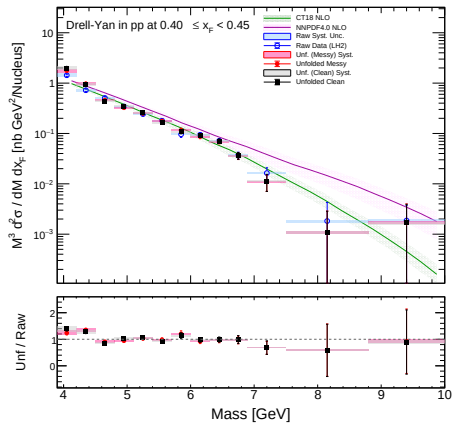


LH2 Target: $0.40 \leq x_F < 0.45$

Iterations = 3

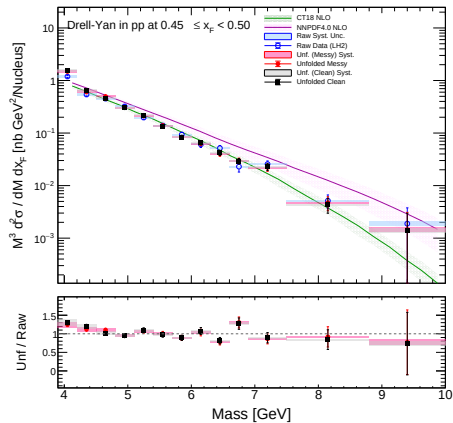


Iterations = 4

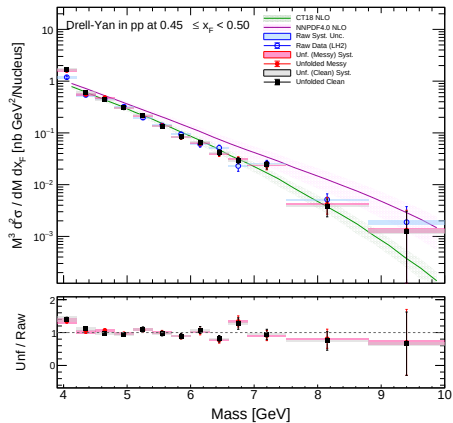


LH2 Target: $0.45 \leq x_F < 0.50$

Iterations = 3

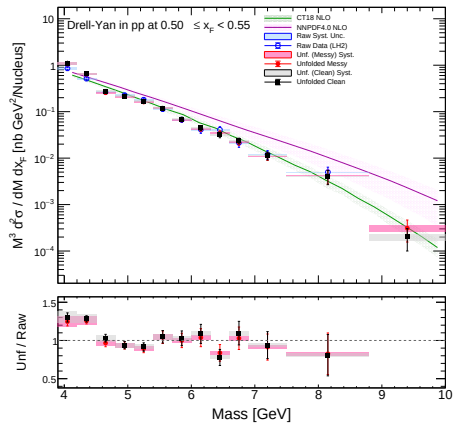


Iterations = 4

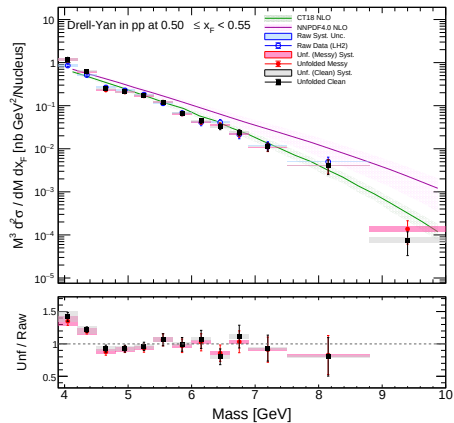


LH2 Target: $0.50 \leq x_F < 0.55$

Iterations = 3

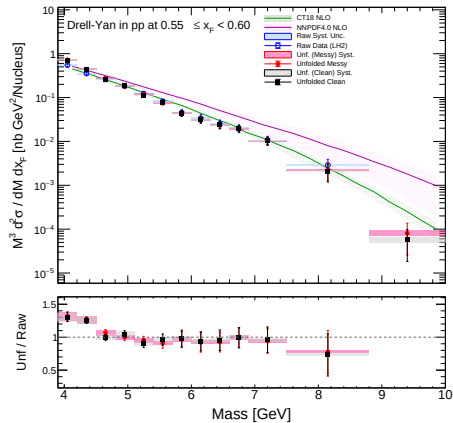


Iterations = 4

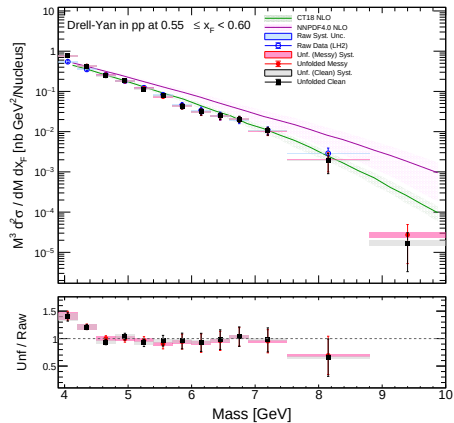


LH2 Target: $0.55 \leq x_F < 0.60$

Iterations = 3

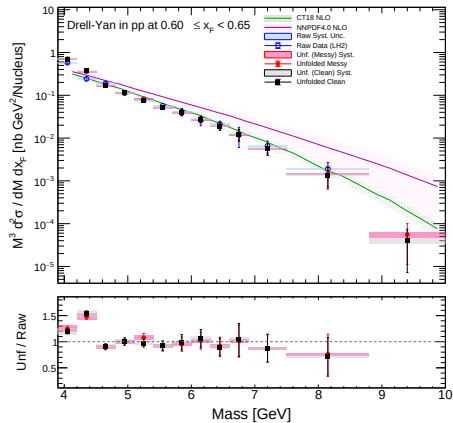


Iterations = 4

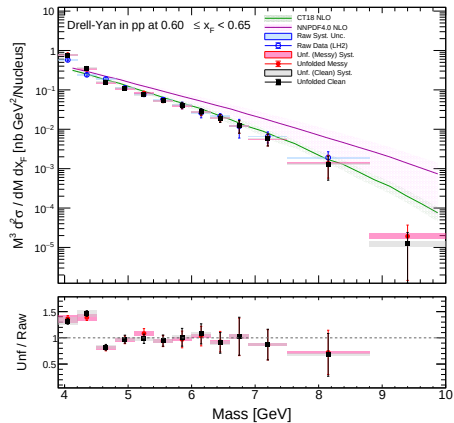


LH2 Target: $0.60 \leq x_F < 0.65$

Iterations = 3

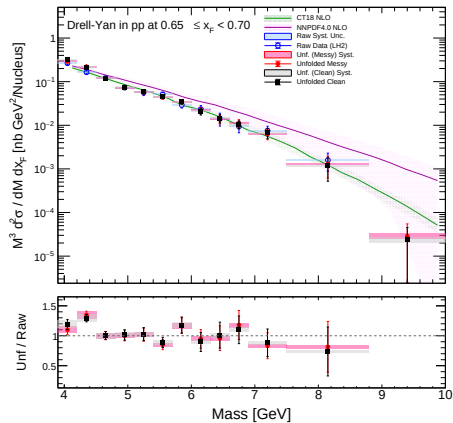


Iterations = 4

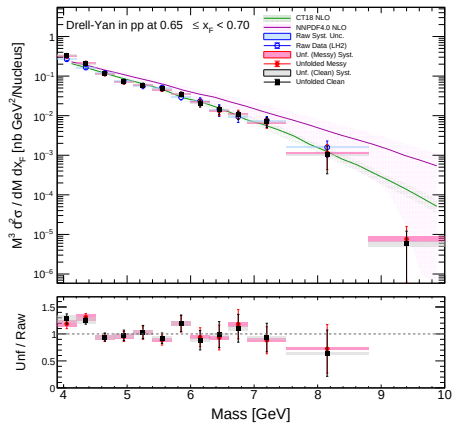


LH2 Target: $0.65 \leq x_F < 0.70$

Iterations = 3

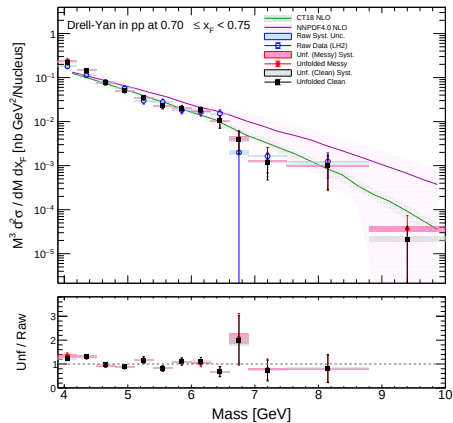


Iterations = 4

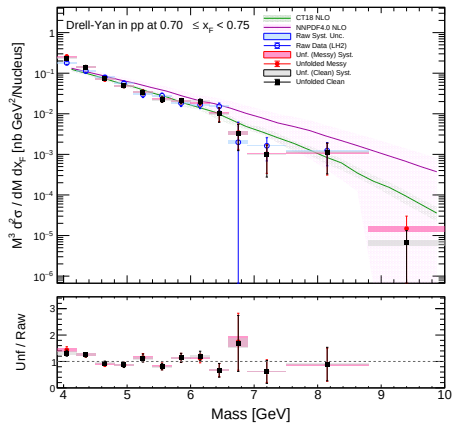


LH2 Target: $0.70 \leq x_F < 0.75$

Iterations = 3

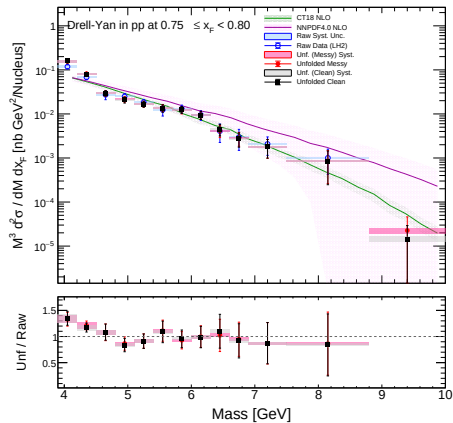


Iterations = 4

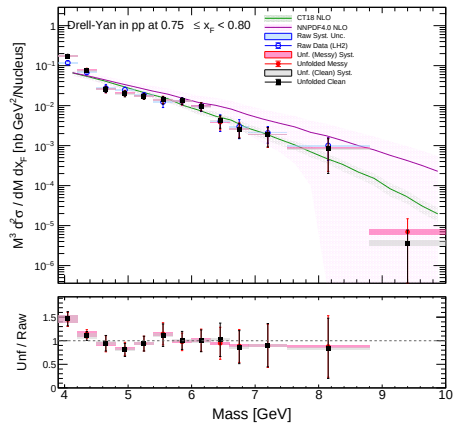


LH2 Target: $0.75 \leq x_F < 0.80$

Iterations = 3

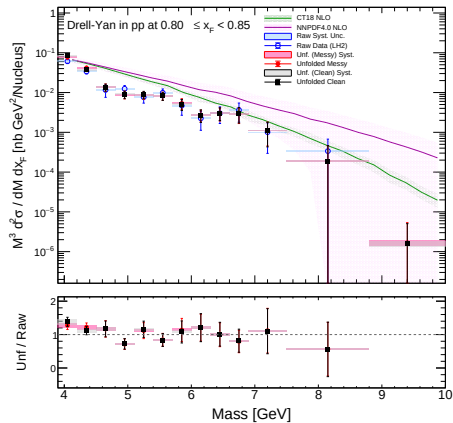


Iterations = 4

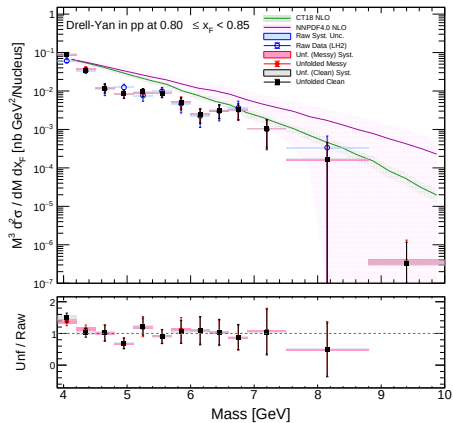


LH2 Target: $0.80 \leq x_F < 0.85$

Iterations = 3

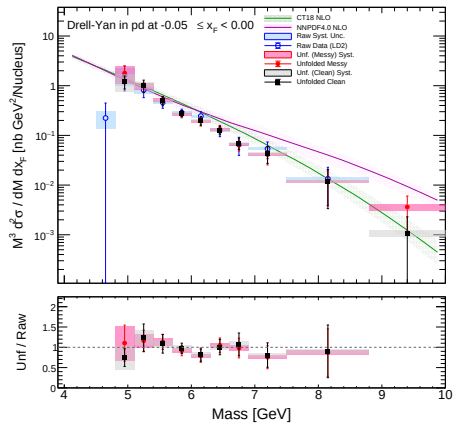


Iterations = 4

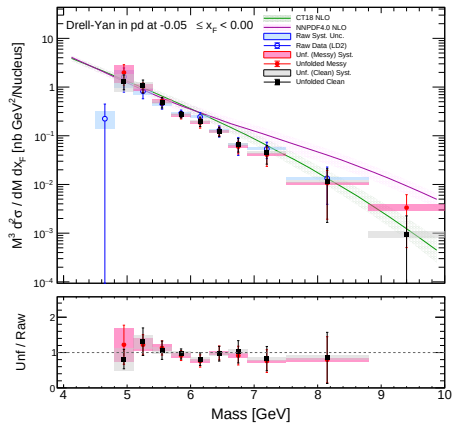


LD2 Target: $-0.05 \leq x_F < 0.00$

Iterations = 3

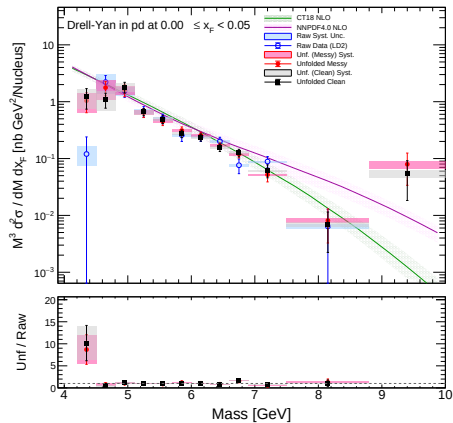


Iterations = 4

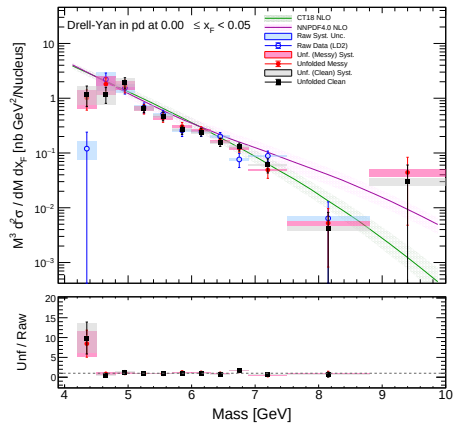


LD2 Target: $0.00 \leq x_F < 0.05$

Iterations = 3

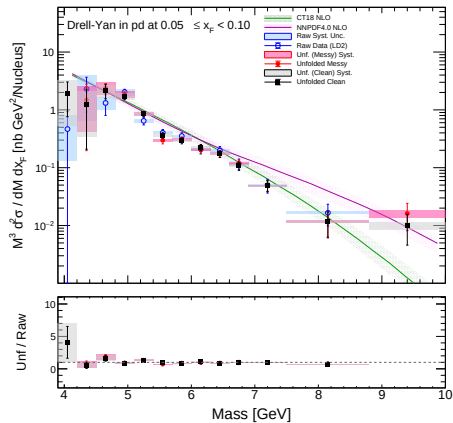


Iterations = 4

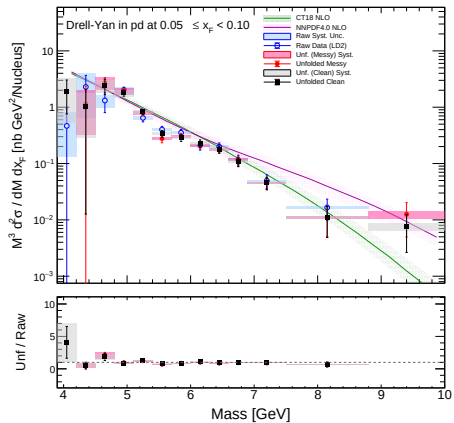


LD2 Target: $0.05 \leq x_F < 0.10$

Iterations = 3

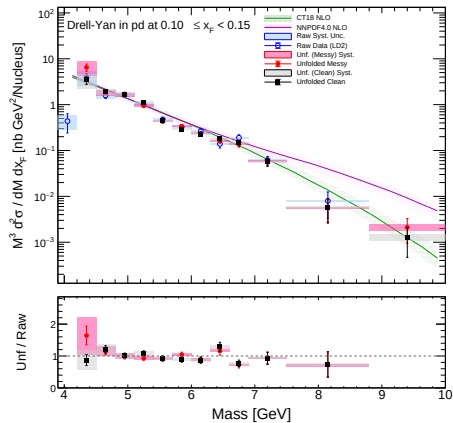


Iterations = 4

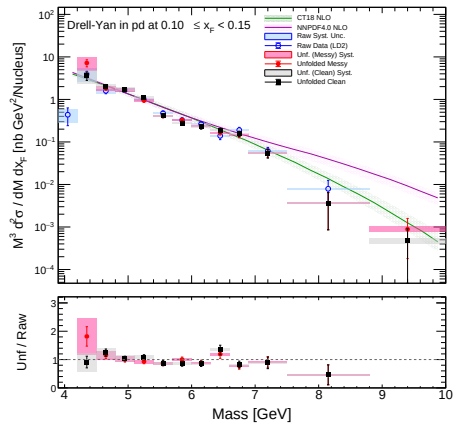


LD2 Target: $0.10 \leq x_F < 0.15$

Iterations = 3

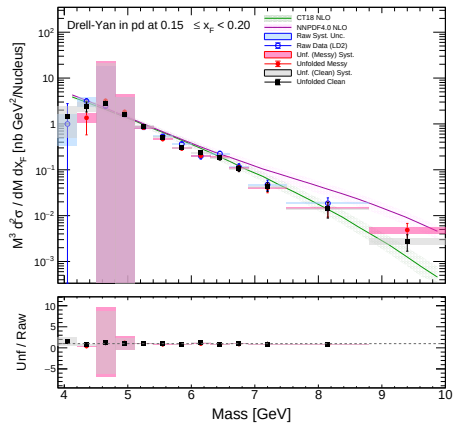


Iterations = 4

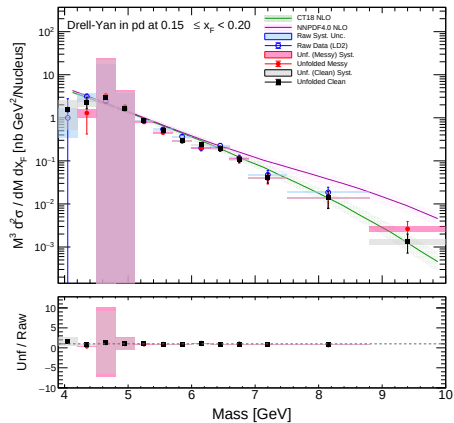


LD2 Target: $0.15 \leq x_F < 0.20$

Iterations = 3

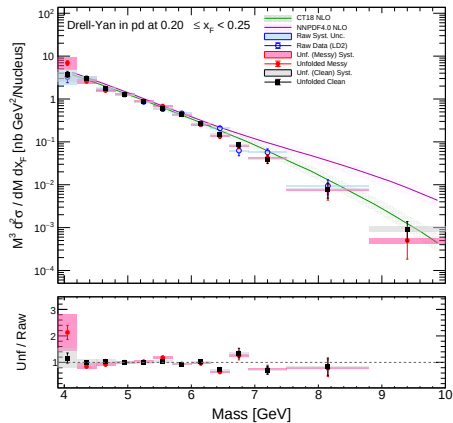


Iterations = 4

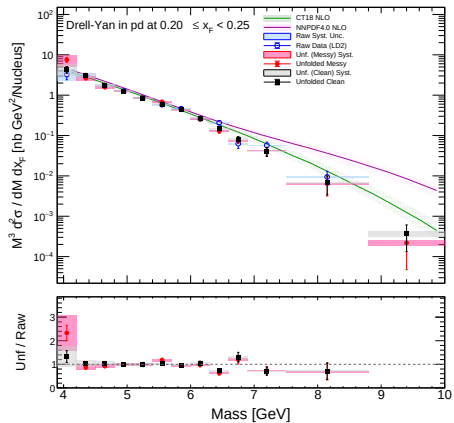


LD2 Target: $0.20 \leq x_F < 0.25$

Iterations = 3

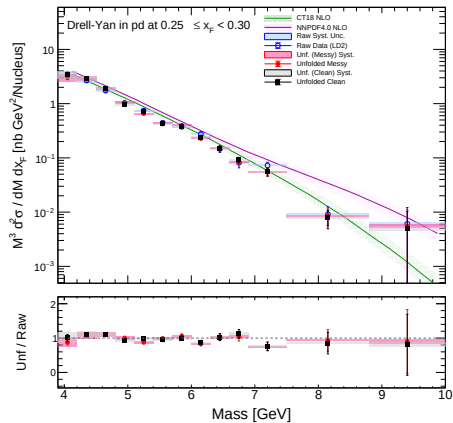


Iterations = 4

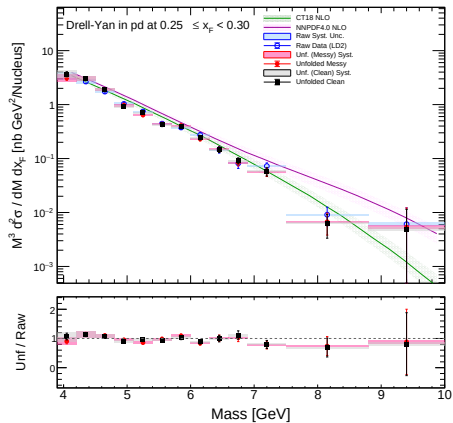


LD2 Target: $0.25 \leq x_F < 0.30$

Iterations = 3

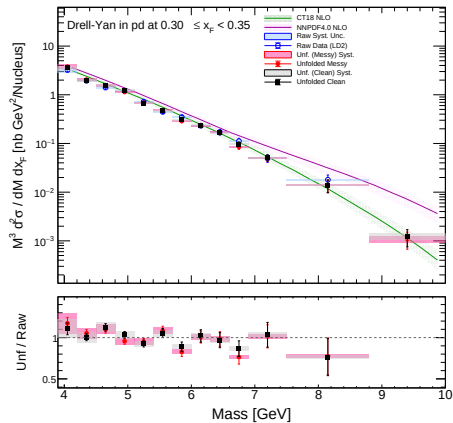


Iterations = 4

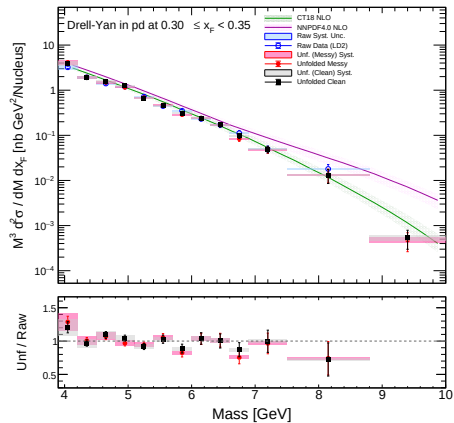


LD2 Target: $0.30 \leq x_F < 0.35$

Iterations = 3

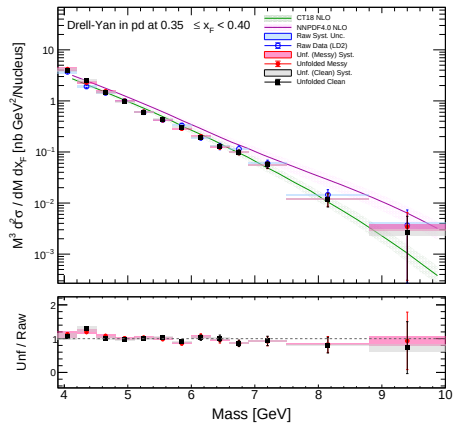


Iterations = 4

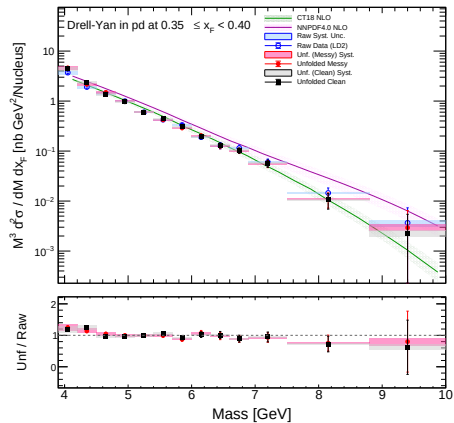


LD2 Target: $0.35 \leq x_F < 0.40$

Iterations = 3

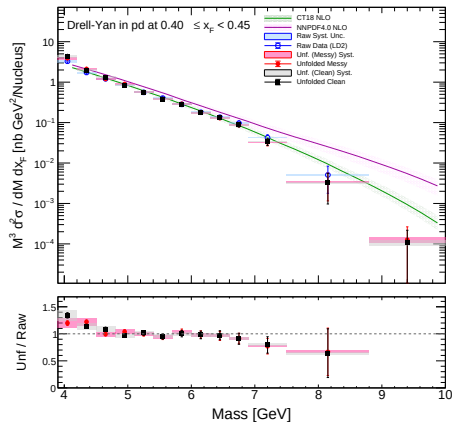


Iterations = 4

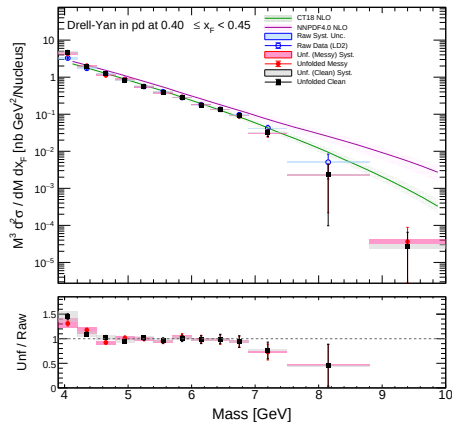


LD2 Target: $0.40 \leq x_F < 0.45$

Iterations = 3

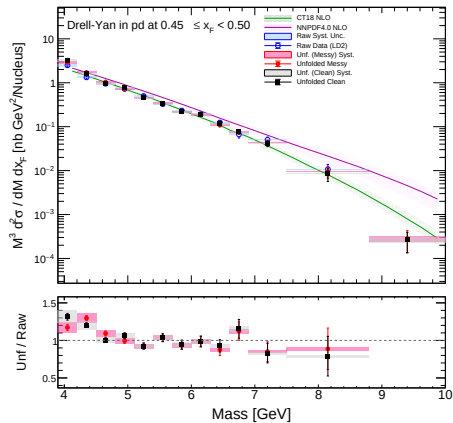


Iterations = 4

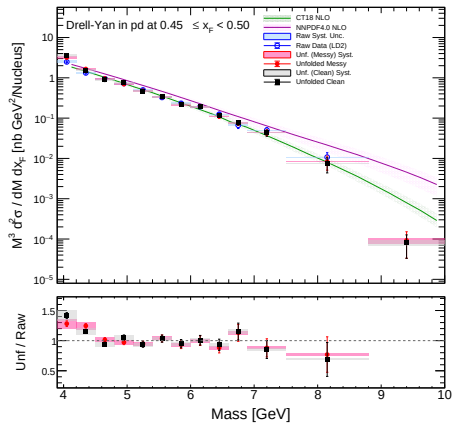


LD2 Target: $0.45 \leq x_F < 0.50$

Iterations = 3

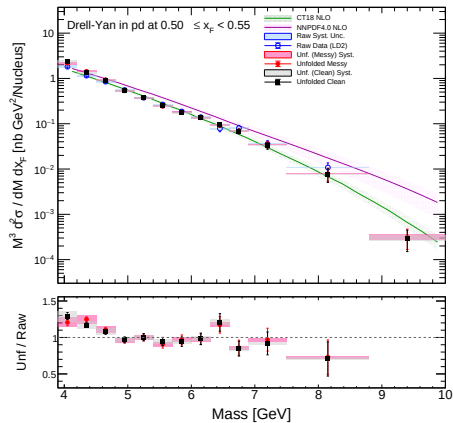


Iterations = 4

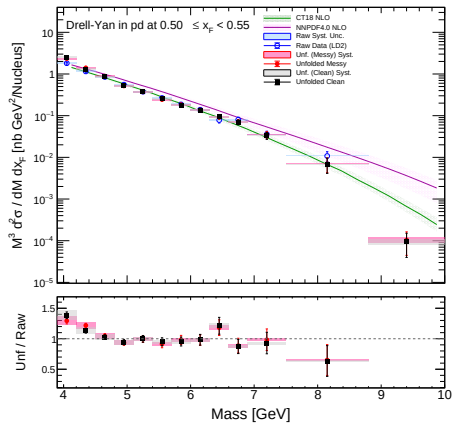


LD2 Target: $0.50 \leq x_F < 0.55$

Iterations = 3

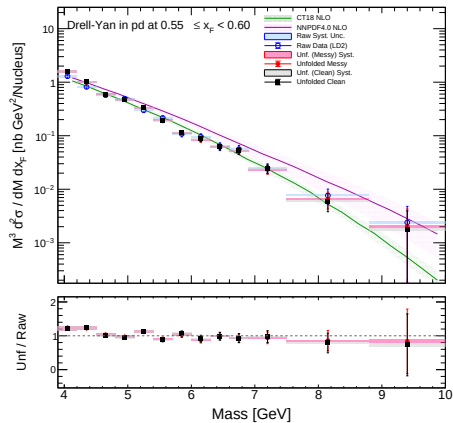


Iterations = 4

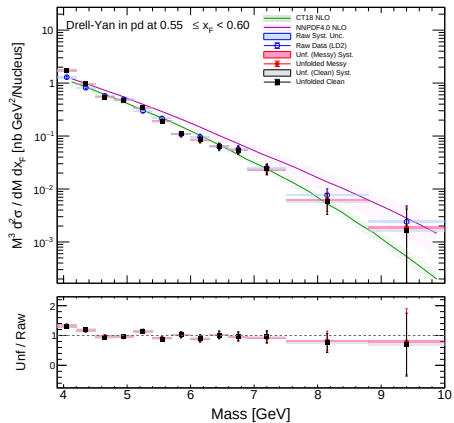


LD2 Target: $0.55 \leq x_F < 0.60$

Iterations = 3

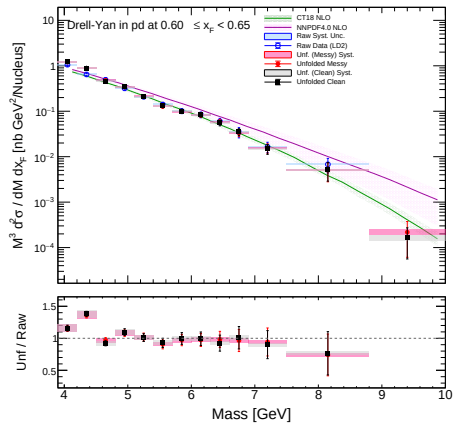


Iterations = 4

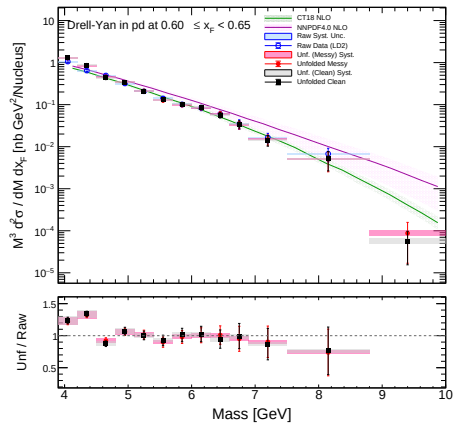


LD2 Target: $0.60 \leq x_F < 0.65$

Iterations = 3

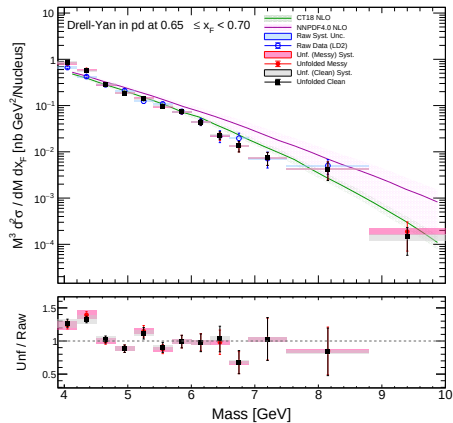


Iterations = 4

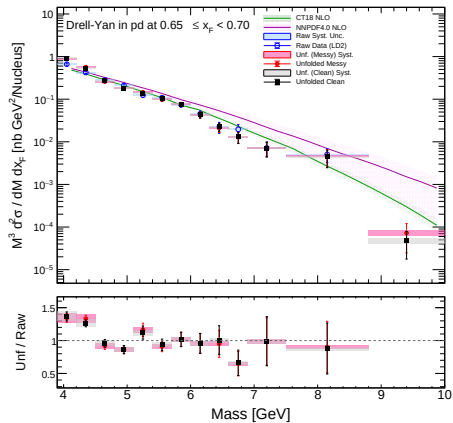


LD2 Target: $0.65 \leq x_F < 0.70$

Iterations = 3

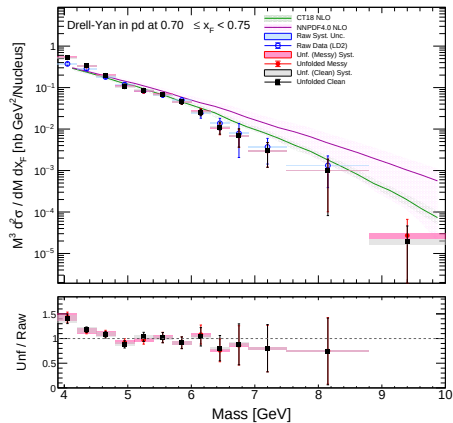


Iterations = 4

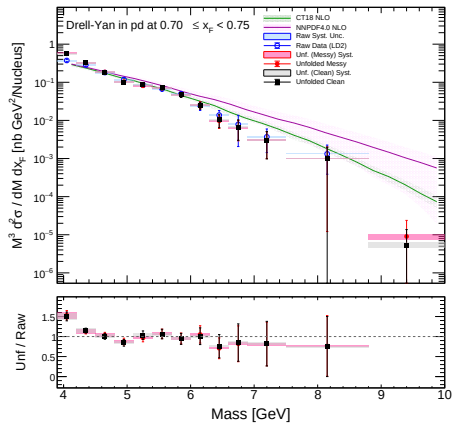


LD2 Target: $0.70 \leq x_F < 0.75$

Iterations = 3

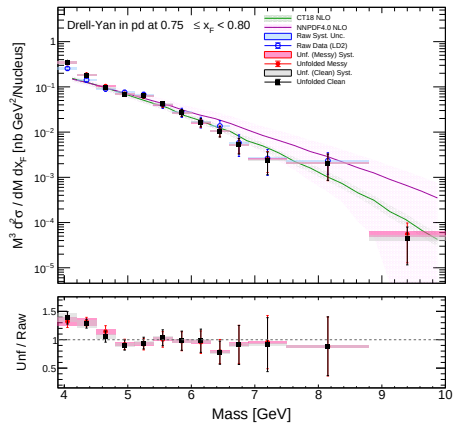


Iterations = 4

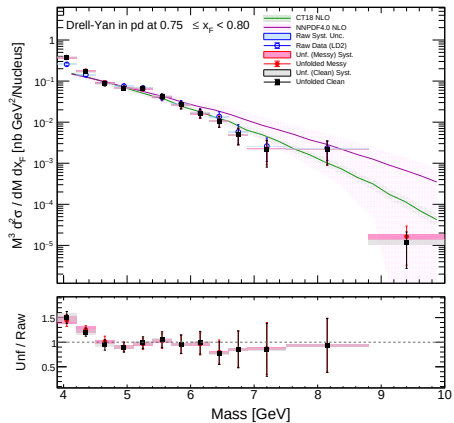


LD2 Target: $0.75 \leq x_F < 0.80$

Iterations = 3

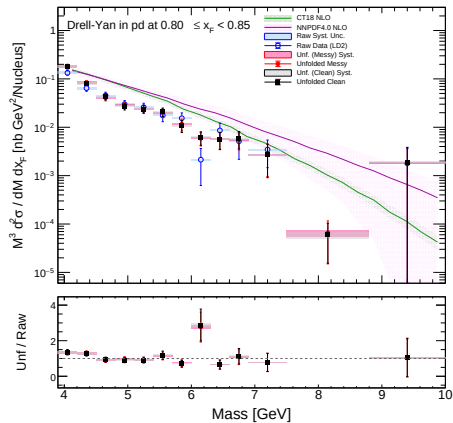


Iterations = 4

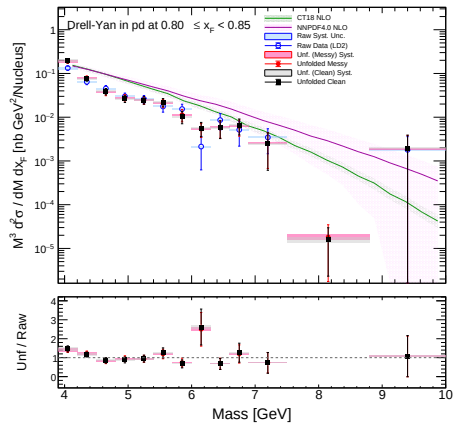


LD2 Target: $0.80 \leq x_F < 0.85$

Iterations = 3



Iterations = 4



Evaluation of Systematic Uncertainty Due to Unfolding

We determine the systematic uncertainty introduced by the unfolding algorithm for both the Messy and Clean configurations.

Systematic Uncertainty Calculation

$$\text{Systematic Uncertainty (\%)} = \left(\frac{|\sigma_{\text{unf}}^{N=4} - \sigma_{\text{unf}}^{N=3}|}{\sigma_{\text{unf}}^{N=3}} \right) \times 100$$

Note: The deviation between consecutive algorithmic iterations provides a robust estimate of the structural bias introduced by the unfolding matrix.

Unfolding Systematics: $-0.05 \leq x_F < 0.00$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.50 \leq \text{Mass} < 4.80$	-	-
$4.80 \leq \text{Mass} < 5.10$	3.55	3.75
$5.10 \leq \text{Mass} < 5.40$	5.51	2.77
$5.40 \leq \text{Mass} < 5.70$	0.66	2.57
$5.70 \leq \text{Mass} < 6.00$	2.82	0.05
$6.00 \leq \text{Mass} < 6.30$	6.88	7.88
$6.30 \leq \text{Mass} < 6.60$	1.48	3.55
$6.60 \leq \text{Mass} < 6.90$	3.75	4.40
$6.90 \leq \text{Mass} < 7.50$	3.06	0.38
$7.50 \leq \text{Mass} < 8.80$	25.12	25.98

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.50 \leq \text{Mass} < 4.80$	-	-
$4.80 \leq \text{Mass} < 5.10$	9.31	10.70
$5.10 \leq \text{Mass} < 5.40$	6.00	4.97
$5.40 \leq \text{Mass} < 5.70$	2.06	0.03
$5.70 \leq \text{Mass} < 6.00$	1.48	0.48
$6.00 \leq \text{Mass} < 6.30$	0.81	3.34
$6.30 \leq \text{Mass} < 6.60$	3.88	5.63
$6.60 \leq \text{Mass} < 6.90$	4.62	6.99
$6.90 \leq \text{Mass} < 7.50$	2.37	0.43
$7.50 \leq \text{Mass} < 8.80$	5.79	8.38

Unfolding Systematics: $0.00 \leq x_F < 0.05$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.20 \leq \text{Mass} < 4.50$	-	-
$4.50 \leq \text{Mass} < 4.80$	16.49	-
$4.80 \leq \text{Mass} < 5.10$	5.66	6.63
$5.10 \leq \text{Mass} < 5.40$	2.81	0.37
$5.40 \leq \text{Mass} < 5.70$	6.20	4.78
$5.70 \leq \text{Mass} < 6.00$	2.41	2.58
$6.00 \leq \text{Mass} < 6.30$	1.76	0.99
$6.30 \leq \text{Mass} < 6.60$	0.96	2.02
$6.60 \leq \text{Mass} < 6.90$	1.80	2.41
$6.90 \leq \text{Mass} < 7.50$	2.45	0.43
$7.50 \leq \text{Mass} < 8.80$	18.71	18.03

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.20 \leq \text{Mass} < 4.50$	2.73	2.68
$4.50 \leq \text{Mass} < 4.80$	9.75	2.46
$4.80 \leq \text{Mass} < 5.10$	6.64	4.62
$5.10 \leq \text{Mass} < 5.40$	1.91	0.92
$5.40 \leq \text{Mass} < 5.70$	3.41	2.22
$5.70 \leq \text{Mass} < 6.00$	3.67	2.47
$6.00 \leq \text{Mass} < 6.30$	1.97	2.49
$6.30 \leq \text{Mass} < 6.60$	1.07	1.04
$6.60 \leq \text{Mass} < 6.90$	3.16	2.40
$6.90 \leq \text{Mass} < 7.50$	1.44	5.79
$7.50 \leq \text{Mass} < 8.80$	38.96	35.50

Unfolding Systematics: $0.05 \leq x_F < 0.10$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.20 \leq \text{Mass} < 4.50$	3.28	2.01
$4.50 \leq \text{Mass} < 4.80$	6.03	5.18
$4.80 \leq \text{Mass} < 5.10$	5.05	6.55
$5.10 \leq \text{Mass} < 5.40$	2.46	0.31
$5.40 \leq \text{Mass} < 5.70$	4.48	4.11
$5.70 \leq \text{Mass} < 6.00$	1.58	0.19
$6.00 \leq \text{Mass} < 6.30$	3.25	3.94
$6.30 \leq \text{Mass} < 6.60$	4.19	4.86
$6.60 \leq \text{Mass} < 6.90$	10.33	9.69
$6.90 \leq \text{Mass} < 7.50$	4.77	6.04
$7.50 \leq \text{Mass} < 8.80$	7.34	7.81

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	0.00	-
$4.20 \leq \text{Mass} < 4.50$	16.81	22.73
$4.50 \leq \text{Mass} < 4.80$	11.53	15.91
$4.80 \leq \text{Mass} < 5.10$	5.84	5.68
$5.10 \leq \text{Mass} < 5.40$	2.89	6.38
$5.40 \leq \text{Mass} < 5.70$	7.01	7.02
$5.70 \leq \text{Mass} < 6.00$	2.67	0.40
$6.00 \leq \text{Mass} < 6.30$	2.98	4.39
$6.30 \leq \text{Mass} < 6.60$	3.87	2.32
$6.60 \leq \text{Mass} < 6.90$	0.53	2.39
$6.90 \leq \text{Mass} < 7.50$	7.07	7.49
$7.50 \leq \text{Mass} < 8.80$	5.06	5.58

Unfolding Systematics: $0.10 \leq x_F < 0.15$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	-	-
$4.20 \leq \text{Mass} < 4.50$	8.54	5.39
$4.50 \leq \text{Mass} < 4.80$	3.55	6.83
$4.80 \leq \text{Mass} < 5.10$	2.20	2.09
$5.10 \leq \text{Mass} < 5.40$	2.78	2.70
$5.40 \leq \text{Mass} < 5.70$	4.26	4.41
$5.70 \leq \text{Mass} < 6.00$	0.11	0.33
$6.00 \leq \text{Mass} < 6.30$	2.21	1.42
$6.30 \leq \text{Mass} < 6.60$	1.33	0.85
$6.60 \leq \text{Mass} < 6.90$	2.17	2.89
$6.90 \leq \text{Mass} < 7.50$	2.40	2.90
$7.50 \leq \text{Mass} < 8.80$	0.29	0.71

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	-	-
$4.20 \leq \text{Mass} < 4.50$	4.42	10.66
$4.50 \leq \text{Mass} < 4.80$	3.01	0.56
$4.80 \leq \text{Mass} < 5.10$	2.20	1.55
$5.10 \leq \text{Mass} < 5.40$	0.06	0.03
$5.40 \leq \text{Mass} < 5.70$	5.87	5.31
$5.70 \leq \text{Mass} < 6.00$	4.06	3.70
$6.00 \leq \text{Mass} < 6.30$	0.66	1.72
$6.30 \leq \text{Mass} < 6.60$	4.06	2.87
$6.60 \leq \text{Mass} < 6.90$	7.21	7.43
$6.90 \leq \text{Mass} < 7.50$	4.43	4.51
$7.50 \leq \text{Mass} < 8.80$	37.79	35.12

Unfolding Systematics: $0.15 \leq x_F < 0.20$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$4.20 \leq \text{Mass} < 4.50$	1.78	1.94
$4.50 \leq \text{Mass} < 4.80$	3.55	2.95
$4.80 \leq \text{Mass} < 5.10$	0.84	0.93
$5.10 \leq \text{Mass} < 5.40$	3.04	1.53
$5.40 \leq \text{Mass} < 5.70$	2.65	2.00
$5.70 \leq \text{Mass} < 6.00$	0.93	0.20
$6.00 \leq \text{Mass} < 6.30$	0.31	2.77
$6.30 \leq \text{Mass} < 6.60$	3.56	3.91
$6.60 \leq \text{Mass} < 6.90$	0.67	1.77
$6.90 \leq \text{Mass} < 7.50$	4.11	3.59
$7.50 \leq \text{Mass} < 8.80$	29.95	26.68
$8.80 \leq \text{Mass} < 10.00$	14.81	20.32

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.52	-
$4.20 \leq \text{Mass} < 4.50$	4.09	5.22
$4.50 \leq \text{Mass} < 4.80$	5.84	5.34
$4.80 \leq \text{Mass} < 5.10$	0.73	0.43
$5.10 \leq \text{Mass} < 5.40$	3.36	3.14
$5.40 \leq \text{Mass} < 5.70$	2.07	3.09
$5.70 \leq \text{Mass} < 6.00$	2.07	2.80
$6.00 \leq \text{Mass} < 6.30$	1.21	0.54
$6.30 \leq \text{Mass} < 6.60$	3.76	5.28
$6.60 \leq \text{Mass} < 6.90$	0.86	1.79
$6.90 \leq \text{Mass} < 7.50$	4.92	3.67
$7.50 \leq \text{Mass} < 8.80$	2.78	4.40

Unfolding Systematics: $0.20 \leq x_F < 0.25$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	5.87	20.21
$4.20 \leq \text{Mass} < 4.50$	8.16	6.27
$4.50 \leq \text{Mass} < 4.80$	3.58	2.42
$4.80 \leq \text{Mass} < 5.10$	2.17	0.33
$5.10 \leq \text{Mass} < 5.40$	0.59	0.12
$5.40 \leq \text{Mass} < 5.70$	0.54	0.77
$5.70 \leq \text{Mass} < 6.00$	1.97	2.54
$6.00 \leq \text{Mass} < 6.30$	1.57	2.02
$6.30 \leq \text{Mass} < 6.60$	2.38	1.26
$6.60 \leq \text{Mass} < 6.90$	3.37	2.33
$6.90 \leq \text{Mass} < 7.50$	4.14	5.79
$7.50 \leq \text{Mass} < 8.80$	0.01	1.77

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	13.87	9.26
$4.20 \leq \text{Mass} < 4.50$	3.32	2.14
$4.50 \leq \text{Mass} < 4.80$	0.06	0.03
$4.80 \leq \text{Mass} < 5.10$	1.78	2.11
$5.10 \leq \text{Mass} < 5.40$	2.46	1.97
$5.40 \leq \text{Mass} < 5.70$	1.05	0.18
$5.70 \leq \text{Mass} < 6.00$	1.34	2.31
$6.00 \leq \text{Mass} < 6.30$	1.69	0.54
$6.30 \leq \text{Mass} < 6.60$	1.54	3.38
$6.60 \leq \text{Mass} < 6.90$	3.57	5.20
$6.90 \leq \text{Mass} < 7.50$	0.99	1.26
$7.50 \leq \text{Mass} < 8.80$	15.43	12.24

Unfolding Systematics: $0.25 \leq x_F < 0.30$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	4.49	1.59
$4.20 \leq \text{Mass} < 4.50$	0.46	0.56
$4.50 \leq \text{Mass} < 4.80$	0.83	3.21
$4.80 \leq \text{Mass} < 5.10$	0.57	0.64
$5.10 \leq \text{Mass} < 5.40$	2.64	4.07
$5.40 \leq \text{Mass} < 5.70$	1.72	2.27
$5.70 \leq \text{Mass} < 6.00$	0.17	0.10
$6.00 \leq \text{Mass} < 6.30$	0.25	0.67
$6.30 \leq \text{Mass} < 6.60$	0.17	0.13
$6.60 \leq \text{Mass} < 6.90$	4.58	3.91
$6.90 \leq \text{Mass} < 7.50$	1.53	2.03
$7.50 \leq \text{Mass} < 8.80$	41.29	38.30

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	6.40	2.92
$4.20 \leq \text{Mass} < 4.50$	2.37	5.60
$4.50 \leq \text{Mass} < 4.80$	1.22	1.13
$4.80 \leq \text{Mass} < 5.10$	3.52	4.97
$5.10 \leq \text{Mass} < 5.40$	2.28	2.34
$5.40 \leq \text{Mass} < 5.70$	0.69	0.38
$5.70 \leq \text{Mass} < 6.00$	2.68	2.47
$6.00 \leq \text{Mass} < 6.30$	1.71	0.82
$6.30 \leq \text{Mass} < 6.60$	3.02	1.90
$6.60 \leq \text{Mass} < 6.90$	0.43	0.78
$6.90 \leq \text{Mass} < 7.50$	3.82	3.96
$7.50 \leq \text{Mass} < 8.80$	19.53	20.18
$8.80 \leq \text{Mass} < 10.00$	0.04	1.99

Unfolding Systematics: $0.30 \leq x_F < 0.35$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	10.70	10.03
$4.20 \leq \text{Mass} < 4.50$	1.99	0.26
$4.50 \leq \text{Mass} < 4.80$	4.08	5.68
$4.80 \leq \text{Mass} < 5.10$	1.08	2.48
$5.10 \leq \text{Mass} < 5.40$	0.79	1.12
$5.40 \leq \text{Mass} < 5.70$	0.52	0.41
$5.70 \leq \text{Mass} < 6.00$	2.05	2.25
$6.00 \leq \text{Mass} < 6.30$	0.62	0.38
$6.30 \leq \text{Mass} < 6.60$	4.00	3.98
$6.60 \leq \text{Mass} < 6.90$	2.01	2.60
$6.90 \leq \text{Mass} < 7.50$	3.94	4.37
$7.50 \leq \text{Mass} < 8.80$	0.29	0.86
$8.80 \leq \text{Mass} < 10.00$	12.68	9.95

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	9.10	9.84
$4.20 \leq \text{Mass} < 4.50$	3.39	4.39
$4.50 \leq \text{Mass} < 4.80$	2.91	2.61
$4.80 \leq \text{Mass} < 5.10$	0.95	1.15
$5.10 \leq \text{Mass} < 5.40$	0.95	0.76
$5.40 \leq \text{Mass} < 5.70$	2.58	2.94
$5.70 \leq \text{Mass} < 6.00$	0.16	0.30
$6.00 \leq \text{Mass} < 6.30$	1.88	2.55
$6.30 \leq \text{Mass} < 6.60$	3.39	3.09
$6.60 \leq \text{Mass} < 6.90$	0.48	1.26
$6.90 \leq \text{Mass} < 7.50$	3.60	5.02
$7.50 \leq \text{Mass} < 8.80$	5.13	3.97

Unfolding Systematics: $0.35 \leq x_F < 0.40$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.16	12.90
$4.20 \leq \text{Mass} < 4.50$	2.81	3.10
$4.50 \leq \text{Mass} < 4.80$	2.72	4.98
$4.80 \leq \text{Mass} < 5.10$	0.56	0.20
$5.10 \leq \text{Mass} < 5.40$	2.11	1.01
$5.40 \leq \text{Mass} < 5.70$	2.44	0.89
$5.70 \leq \text{Mass} < 6.00$	2.07	0.84
$6.00 \leq \text{Mass} < 6.30$	1.72	2.08
$6.30 \leq \text{Mass} < 6.60$	0.82	1.00
$6.60 \leq \text{Mass} < 6.90$	0.85	1.50
$6.90 \leq \text{Mass} < 7.50$	1.05	2.12
$7.50 \leq \text{Mass} < 8.80$	1.74	1.47

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	10.00	10.35
$4.20 \leq \text{Mass} < 4.50$	3.87	5.10
$4.50 \leq \text{Mass} < 4.80$	4.68	3.20
$4.80 \leq \text{Mass} < 5.10$	1.09	1.37
$5.10 \leq \text{Mass} < 5.40$	1.20	2.41
$5.40 \leq \text{Mass} < 5.70$	0.56	0.40
$5.70 \leq \text{Mass} < 6.00$	1.42	1.29
$6.00 \leq \text{Mass} < 6.30$	0.70	0.10
$6.30 \leq \text{Mass} < 6.60$	0.18	0.94
$6.60 \leq \text{Mass} < 6.90$	3.91	3.31
$6.90 \leq \text{Mass} < 7.50$	1.59	1.10
$7.50 \leq \text{Mass} < 8.80$	10.52	9.12
$8.80 \leq \text{Mass} < 10.00$	15.55	14.46

Unfolding Systematics: $0.40 \leq x_F < 0.45$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	9.02	10.93
$4.20 \leq \text{Mass} < 4.50$	4.39	4.05
$4.50 \leq \text{Mass} < 4.80$	6.52	7.53
$4.80 \leq \text{Mass} < 5.10$	0.74	1.79
$5.10 \leq \text{Mass} < 5.40$	2.02	2.35
$5.40 \leq \text{Mass} < 5.70$	0.16	0.81
$5.70 \leq \text{Mass} < 6.00$	2.07	1.11
$6.00 \leq \text{Mass} < 6.30$	2.35	3.08
$6.30 \leq \text{Mass} < 6.60$	4.19	3.61
$6.60 \leq \text{Mass} < 6.90$	0.23	1.38
$6.90 \leq \text{Mass} < 7.50$	8.66	10.60
$7.50 \leq \text{Mass} < 8.80$	24.85	26.33
$8.80 \leq \text{Mass} < 10.00$	23.89	23.36

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.21	9.37
$4.20 \leq \text{Mass} < 4.50$	4.04	3.63
$4.50 \leq \text{Mass} < 4.80$	5.30	7.27
$4.80 \leq \text{Mass} < 5.10$	2.41	2.42
$5.10 \leq \text{Mass} < 5.40$	0.08	0.53
$5.40 \leq \text{Mass} < 5.70$	1.41	1.28
$5.70 \leq \text{Mass} < 6.00$	0.49	0.55
$6.00 \leq \text{Mass} < 6.30$	0.09	0.64
$6.30 \leq \text{Mass} < 6.60$	2.12	2.23
$6.60 \leq \text{Mass} < 6.90$	3.63	2.91
$6.90 \leq \text{Mass} < 7.50$	4.24	5.41
$7.50 \leq \text{Mass} < 8.80$	29.36	30.42

Unfolding Systematics: $0.45 \leq x_F < 0.50$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.01	7.89
$4.20 \leq \text{Mass} < 4.50$	7.29	8.44
$4.50 \leq \text{Mass} < 4.80$	3.95	3.16
$4.80 \leq \text{Mass} < 5.10$	0.00	0.68
$5.10 \leq \text{Mass} < 5.40$	0.30	0.48
$5.40 \leq \text{Mass} < 5.70$	0.23	0.75
$5.70 \leq \text{Mass} < 6.00$	0.01	0.87
$6.00 \leq \text{Mass} < 6.30$	0.31	0.19
$6.30 \leq \text{Mass} < 6.60$	0.33	1.15
$6.60 \leq \text{Mass} < 6.90$	0.02	2.48
$6.90 \leq \text{Mass} < 7.50$	5.18	5.39
$7.50 \leq \text{Mass} < 8.80$	11.65	11.48
$8.80 \leq \text{Mass} < 10.00$	10.11	9.01

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.63	9.24
$4.20 \leq \text{Mass} < 4.50$	4.33	3.94
$4.50 \leq \text{Mass} < 4.80$	6.54	7.21
$4.80 \leq \text{Mass} < 5.10$	1.09	2.32
$5.10 \leq \text{Mass} < 5.40$	1.13	2.37
$5.40 \leq \text{Mass} < 5.70$	0.14	0.02
$5.70 \leq \text{Mass} < 6.00$	0.61	0.43
$6.00 \leq \text{Mass} < 6.30$	1.97	1.76
$6.30 \leq \text{Mass} < 6.60$	0.46	0.76
$6.60 \leq \text{Mass} < 6.90$	0.31	0.39
$6.90 \leq \text{Mass} < 7.50$	2.58	3.51
$7.50 \leq \text{Mass} < 8.80$	12.96	13.39

Unfolding Systematics: $0.50 \leq x_F < 0.55$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.94	8.32
$4.20 \leq \text{Mass} < 4.50$	4.82	4.90
$4.50 \leq \text{Mass} < 4.80$	9.91	9.26
$4.80 \leq \text{Mass} < 5.10$	1.26	1.71
$5.10 \leq \text{Mass} < 5.40$	5.27	4.48
$5.40 \leq \text{Mass} < 5.70$	1.51	1.98
$5.70 \leq \text{Mass} < 6.00$	3.50	2.93
$6.00 \leq \text{Mass} < 6.30$	1.49	1.20
$6.30 \leq \text{Mass} < 6.60$	3.04	2.32
$6.60 \leq \text{Mass} < 6.90$	1.55	0.38
$6.90 \leq \text{Mass} < 7.50$	0.45	0.35
$7.50 \leq \text{Mass} < 8.80$	0.95	0.24

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.02	7.34
$4.20 \leq \text{Mass} < 4.50$	3.65	2.51
$4.50 \leq \text{Mass} < 4.80$	4.66	5.33
$4.80 \leq \text{Mass} < 5.10$	2.38	2.72
$5.10 \leq \text{Mass} < 5.40$	0.56	0.16
$5.40 \leq \text{Mass} < 5.70$	0.66	0.59
$5.70 \leq \text{Mass} < 6.00$	1.33	1.13
$6.00 \leq \text{Mass} < 6.30$	0.16	0.24
$6.30 \leq \text{Mass} < 6.60$	0.52	0.78
$6.60 \leq \text{Mass} < 6.90$	3.15	3.24
$6.90 \leq \text{Mass} < 7.50$	1.25	1.38
$7.50 \leq \text{Mass} < 8.80$	9.95	10.87

Unfolding Systematics: $0.55 \leq x_F < 0.60$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	6.77	7.69
$4.20 \leq \text{Mass} < 4.50$	3.47	3.87
$4.50 \leq \text{Mass} < 4.80$	6.09	6.20
$4.80 \leq \text{Mass} < 5.10$	0.76	0.51
$5.10 \leq \text{Mass} < 5.40$	1.81	1.44
$5.40 \leq \text{Mass} < 5.70$	0.33	0.93
$5.70 \leq \text{Mass} < 6.00$	2.47	2.15
$6.00 \leq \text{Mass} < 6.30$	0.62	0.31
$6.30 \leq \text{Mass} < 6.60$	2.90	3.36
$6.60 \leq \text{Mass} < 6.90$	4.17	4.39
$6.90 \leq \text{Mass} < 7.50$	1.45	1.06
$7.50 \leq \text{Mass} < 8.80$	10.58	10.67

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.28	7.54
$4.20 \leq \text{Mass} < 4.50$	4.30	4.49
$4.50 \leq \text{Mass} < 4.80$	6.29	5.86
$4.80 \leq \text{Mass} < 5.10$	0.19	0.92
$5.10 \leq \text{Mass} < 5.40$	1.92	1.35
$5.40 \leq \text{Mass} < 5.70$	0.54	0.19
$5.70 \leq \text{Mass} < 6.00$	2.56	2.12
$6.00 \leq \text{Mass} < 6.30$	0.43	0.54
$6.30 \leq \text{Mass} < 6.60$	3.24	2.92
$6.60 \leq \text{Mass} < 6.90$	4.07	3.89
$6.90 \leq \text{Mass} < 7.50$	1.24	0.86
$7.50 \leq \text{Mass} < 8.80$	4.79	4.42
$8.80 \leq \text{Mass} < 10.00$	5.31	5.60

Unfolding Systematics: $0.60 \leq x_F < 0.65$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	9.53	9.74
$4.20 \leq \text{Mass} < 4.50$	4.87	6.04
$4.50 \leq \text{Mass} < 4.80$	10.02	9.71
$4.80 \leq \text{Mass} < 5.10$	3.03	3.44
$5.10 \leq \text{Mass} < 5.40$	0.74	0.62
$5.40 \leq \text{Mass} < 5.70$	2.57	2.72
$5.70 \leq \text{Mass} < 6.00$	2.10	1.77
$6.00 \leq \text{Mass} < 6.30$	1.23	1.14
$6.30 \leq \text{Mass} < 6.60$	1.14	1.30
$6.60 \leq \text{Mass} < 6.90$	0.22	0.56
$6.90 \leq \text{Mass} < 7.50$	0.73	0.50
$7.50 \leq \text{Mass} < 8.80$	4.95	4.90

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	6.69	5.91
$4.20 \leq \text{Mass} < 4.50$	3.03	2.47
$4.50 \leq \text{Mass} < 4.80$	5.19	4.92
$4.80 \leq \text{Mass} < 5.10$	2.24	1.77
$5.10 \leq \text{Mass} < 5.40$	0.70	0.29
$5.40 \leq \text{Mass} < 5.70$	0.21	0.72
$5.70 \leq \text{Mass} < 6.00$	1.37	0.76
$6.00 \leq \text{Mass} < 6.30$	3.29	3.37
$6.30 \leq \text{Mass} < 6.60$	2.43	2.68
$6.60 \leq \text{Mass} < 6.90$	1.46	1.44
$6.90 \leq \text{Mass} < 7.50$	3.59	4.00
$7.50 \leq \text{Mass} < 8.80$	0.34	0.32

Unfolding Systematics: $0.65 \leq x_F < 0.70$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.47	7.86
$4.20 \leq \text{Mass} < 4.50$	3.67	2.75
$4.50 \leq \text{Mass} < 4.80$	6.01	7.16
$4.80 \leq \text{Mass} < 5.10$	3.91	4.78
$5.10 \leq \text{Mass} < 5.40$	0.72	0.68
$5.40 \leq \text{Mass} < 5.70$	3.68	4.19
$5.70 \leq \text{Mass} < 6.00$	1.45	1.71
$6.00 \leq \text{Mass} < 6.30$	1.38	0.95
$6.30 \leq \text{Mass} < 6.60$	2.35	2.84
$6.60 \leq \text{Mass} < 6.90$	0.50	0.04
$6.90 \leq \text{Mass} < 7.50$	5.72	5.92
$7.50 \leq \text{Mass} < 8.80$	12.75	11.11

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.64	8.49
$4.20 \leq \text{Mass} < 4.50$	4.75	4.34
$4.50 \leq \text{Mass} < 4.80$	6.46	8.04
$4.80 \leq \text{Mass} < 5.10$	2.35	2.48
$5.10 \leq \text{Mass} < 5.40$	0.32	1.43
$5.40 \leq \text{Mass} < 5.70$	3.51	3.66
$5.70 \leq \text{Mass} < 6.00$	2.75	2.67
$6.00 \leq \text{Mass} < 6.30$	1.71	2.28
$6.30 \leq \text{Mass} < 6.60$	2.11	3.08
$6.60 \leq \text{Mass} < 6.90$	2.80	3.68
$6.90 \leq \text{Mass} < 7.50$	3.87	4.01
$7.50 \leq \text{Mass} < 8.80$	5.15	5.88

Unfolding Systematics: $0.70 \leq x_F < 0.75$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	5.55	6.20
$4.20 \leq \text{Mass} < 4.50$	3.37	4.21
$4.50 \leq \text{Mass} < 4.80$	3.66	4.57
$4.80 \leq \text{Mass} < 5.10$	2.02	1.93
$5.10 \leq \text{Mass} < 5.40$	2.67	1.85
$5.40 \leq \text{Mass} < 5.70$	0.86	0.18
$5.70 \leq \text{Mass} < 6.00$	3.29	3.15
$6.00 \leq \text{Mass} < 6.30$	7.62	7.61
$6.30 \leq \text{Mass} < 6.60$	2.16	1.16
$6.60 \leq \text{Mass} < 6.90$	15.29	15.68
$6.90 \leq \text{Mass} < 7.50$	16.78	17.96
$7.50 \leq \text{Mass} < 8.80$	10.03	11.43

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.27	6.99
$4.20 \leq \text{Mass} < 4.50$	2.55	2.08
$4.50 \leq \text{Mass} < 4.80$	6.61	6.26
$4.80 \leq \text{Mass} < 5.10$	4.63	4.93
$5.10 \leq \text{Mass} < 5.40$	0.27	0.64
$5.40 \leq \text{Mass} < 5.70$	4.44	4.37
$5.70 \leq \text{Mass} < 6.00$	3.01	4.23
$6.00 \leq \text{Mass} < 6.30$	2.60	2.56
$6.30 \leq \text{Mass} < 6.60$	5.70	7.16
$6.60 \leq \text{Mass} < 6.90$	3.69	4.86
$6.90 \leq \text{Mass} < 7.50$	2.27	2.17
$7.50 \leq \text{Mass} < 8.80$	1.63	2.19

Unfolding Systematics: $0.75 \leq x_F < 0.80$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.78	9.16
$4.20 \leq \text{Mass} < 4.50$	6.09	5.54
$4.50 \leq \text{Mass} < 4.80$	12.66	13.38
$4.80 \leq \text{Mass} < 5.10$	3.71	4.40
$5.10 \leq \text{Mass} < 5.40$	2.97	2.87
$5.40 \leq \text{Mass} < 5.70$	2.71	3.19
$5.70 \leq \text{Mass} < 6.00$	5.32	5.11
$6.00 \leq \text{Mass} < 6.30$	0.79	1.29
$6.30 \leq \text{Mass} < 6.60$	7.34	7.41
$6.60 \leq \text{Mass} < 6.90$	5.86	6.14
$6.90 \leq \text{Mass} < 7.50$	3.79	2.37
$7.50 \leq \text{Mass} < 8.80$	0.04	1.53

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	8.43	9.00
$4.20 \leq \text{Mass} < 4.50$	6.03	4.81
$4.50 \leq \text{Mass} < 4.80$	10.66	11.52
$4.80 \leq \text{Mass} < 5.10$	0.93	1.49
$5.10 \leq \text{Mass} < 5.40$	5.41	5.10
$5.40 \leq \text{Mass} < 5.70$	1.71	2.09
$5.70 \leq \text{Mass} < 6.00$	2.27	2.09
$6.00 \leq \text{Mass} < 6.30$	0.66	0.76
$6.30 \leq \text{Mass} < 6.60$	0.64	1.07
$6.60 \leq \text{Mass} < 6.90$	5.78	6.79
$6.90 \leq \text{Mass} < 7.50$	7.78	8.94
$7.50 \leq \text{Mass} < 8.80$	5.45	6.44

Unfolding Systematics: $0.80 \leq x_F < 0.85$

LH2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.88	8.37
$4.20 \leq \text{Mass} < 4.50$	8.83	7.87
$4.50 \leq \text{Mass} < 4.80$	13.07	13.90
$4.80 \leq \text{Mass} < 5.10$	4.26	4.72
$5.10 \leq \text{Mass} < 5.40$	6.51	5.72
$5.40 \leq \text{Mass} < 5.70$	7.01	7.54
$5.70 \leq \text{Mass} < 6.00$	2.76	2.08
$6.00 \leq \text{Mass} < 6.30$	10.21	9.48
$6.30 \leq \text{Mass} < 6.60$	3.20	2.26
$6.60 \leq \text{Mass} < 6.90$	8.58	8.85
$6.90 \leq \text{Mass} < 7.50$	5.07	3.81
$7.50 \leq \text{Mass} < 8.80$	10.41	13.76

LD2 Target

Mass Bin	Syst. Error (Clean) [%]	Syst. Error (Messy) [%]
$3.90 \leq \text{Mass} < 4.20$	7.54	7.32
$4.20 \leq \text{Mass} < 4.50$	7.01	6.35
$4.50 \leq \text{Mass} < 4.80$	8.91	8.68
$4.80 \leq \text{Mass} < 5.10$	2.37	1.97
$5.10 \leq \text{Mass} < 5.40$	3.91	3.41
$5.40 \leq \text{Mass} < 5.70$	5.86	4.57
$5.70 \leq \text{Mass} < 6.00$	3.76	1.25
$6.00 \leq \text{Mass} < 6.30$	9.16	9.74
$6.30 \leq \text{Mass} < 6.60$	2.00	2.66
$6.60 \leq \text{Mass} < 6.90$	12.11	11.44
$6.90 \leq \text{Mass} < 7.50$	6.96	6.34
$8.80 \leq \text{Mass} < 10.00$	2.60	2.59

Conclusions and Next Steps

- Successfully implemented multi-dimensional unfolding for the double differential cross-section.
- Recalculation of acceptance corrections for Runs 2 and 3 independently is currently underway (Kenichi and Harsha).
- Finalize the publication draft and initiate the collaboration review process.